Caption-Image Retrieval

UE24CS352A Machine Learning — Mini-Project

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1. Problem Statement

Given a collection of images and a natural-language query describing a desired image, return a ranked list of the top-K most relevant images. This is content-based image retrieval, framed as learning a *shared embedding space* in which captions and images are aligned, so that nearest-neighbour search for a query's embedding returns the correct image.

2. Dataset

Microsoft COCO (Lin et al., 2014) with the Karpathy & Fei-Fei (2015) split: 113,287 training images, 5,000 validation, 5,000 test, each with five human-annotated captions. Image features are the precomputed 10-crop VGG-19 `fc7` vectors from Vendrov et al. (2015), 4,096 dimensions per image; raw pixels are not used. The baseline represents a caption as the mean of its GloVe (Pennington et al., 2014) 300-dimensional word vectors; the neural model consumes the same vectors through a GRU.

3. Approach

Both models map captions and images into one shared space.

Baseline — multi-response linear regression. A weight matrix `W` is fit so that `X @ W` approximates the image features `Y` in a least-squares sense, where `X` holds mean-pooled GloVe caption vectors and `Y` the aligned image features. Since each image has five captions, each image feature is repeated five times so rows align. The fit is closed form; retrieval projects a query with `W` and returns the nearest image features.

GRU + GloVe. Following Vendrov et al. (2015), an image is projected by a learned linear map `f_i(i) = W_i · f_VGG(i)` and a caption encoded by a GRU initialized from GloVe, `f_c(c) = GRU(c)`. Both encoders are trained with a margin ranking loss: for each correct pair, wrong captions and wrong images in the batch are penalized when their similarity comes within a margin of the positive pair. Similarity is the negative squared Euclidean distance between L2-normalized embeddings.

4. Implementation

Code is in `src/` : NumPy for feature handling, PyTorch for the model, CPU training. `data.py` , `glove.py` , and `vocab.py` load features, captions, and the GloVe embedding matrix; `baseline.py` fits the least-squares `W` and computes Recall@K; `models.py` defines the image and GRU encoders and the margin loss; `train_rnn.py` trains and evaluates; `retrieve.py` and `demo.py` provide interactive query-to-image retrieval.

5. Results

Model	R@1	R@5	R@10	Mean rank
Baseline	2.8	9.9	16.7	179.4
GRU + GloVe	5.0	18.0	28.3	85.6

Vendrov et al. (reported)	37.0	—	86.8	7.3

Our models train on a 5,000-image subset and evaluate on the full 5,000-image validation set. The reported row used the full training set and different feature extraction, so it is not directly comparable.

6. Conclusions

The GRU encoder improves on the baseline on every metric: R@10 rises from 16.7% to 28.3% and the mean rank of the correct image roughly halves, from 179 to 86. The baseline's weakness is structural — averaging a caption's word vectors discards word order and dilutes the object described, so captions naming several objects retrieve poorly, whereas the GRU consumes the words in sequence and can represent their relations. Our absolute numbers are well below the reported 86.8% because we train on a 5,000-image subset for 30 epochs rather than the full 113,287 images for 50; both models were still improving slightly at the final epoch, indicating data scale was the limiting factor. Limitations: a single global image vector with no region attention; precomputed features rather than recomputed; and evaluation on a reduced training split.